Creating Everyone Everywhere Affordable Broadband Access

by Dan and Jamie Douglass
July 24, 2022

Executive Summary

This paper provides essential recommendations for creating affordable fiber optic broadband access. The current goal for service is 100 Mbps symmetrical (download and upload speed) at no more than $70/month. By 2032, 1 Gbps should be available for $100/month. Fiber optic technology should be used and the resulting network infrastructure should be publicly owned and regulated. Neighborhood networking projects decompose the larger network allowing the pursuit of a divide and conquer strategy. Each neighborhood project is driven by local support which can be financed as Local Utility Districts (LUD), grants, or other funding sources. A menu of private Internet Service Provider (ISP) companies compete to provide end-user services. Tools and other considerations that support effective implementation include online network performance testing and mapping to direct network implementation, cost-benefit analysis using total cost of operation for decisions such as underground versus aerial fiber optic installation, and implementation of a dig once policy to accelerate network backbone growth while saving both time and money. Dig once needs proper incentives to reward compliance.

© 2022 Research on Knowing LLC

Introduction

The need for affordable broadband access has been elevated within the concerns facing the public. The internet has become the source most users turn to first for information. During the COVID-19 pandemic, internet access was needed by the public for health, education, and conducting business. Large-scale network implementation to reach everyone despite where they are located is a daunting problem. In this paper, we set forth our recommendations for addressing society's need to create affordable broadband access everywhere and for everyone. There are many important areas to truly achieve the goal of universal broadband access. Having end-user devices such as a mobile phone, tablet, laptop or desktop computer along with the skill of digital literacy remain challenges others are working to address. This paper will focus solely on the broadband infrastructure and Internet services within the United States.

Definition: Affordable.

No more than $70/mo. for 100 Mbps service and $100/mo. for Gbps service. Kitsap PUD offers services that met these affordability requirements. ("Kitsap PUD," 2022) Like other utilities, low-income households should be subsidized including free service for some. These costs for services are based on surveying similar offerings within Washington State by publicly owned internet services providing fiber optic broadband.

Definition: Internet Service Standards.

The Federal Communications Commission (FCC) defines high-speed internet as 25 Mbps download and 3 Mbps upload. (Velazco, 2022) Applying this standard to local residences proclaims many locations as served while users complain about inadequate services. Teleconferencing is only one example where the FCC standard fails to perform and users experience dropped and disrupted service. In 2022, Washington State Senate Bill 5715 defined a new statewide standard of 100 Mbps down and 20 Mbps up. (Bode, 2022) Both standards have low upload speeds. They assume user locations supply only small bursts of data. As applications expand and user locations increasingly become sources of more and more data with smart homes and other

© 2022 Research on Knowing LLC

systems, the need for increased upload speeds will ramp up. Fiber optic networks can accommodate symmetrical transmission speeds which places devices on a pier to pier status rather than the provider-customer relationship of current end-user devices.

Therefore, we propose a current standard of symmetrical 100 Mbps for both download and upload. The growth pattern of internet use is expected to continue exponentially. By 2032, 1 Gbps symmetrical with less than 30 ms latency will be needed to support adequate service for users.

Fiber Optic Infrastructure

Recommendation: Fiber Optic network infrastructure whenever possible.

Fiber optics is a future-proof technology for network implementation. ("Why Fiber-Optic Internet Is a Long Term, Future-Proof Solution | WRAL TechWire," 2021) By the laws of physics, the signal to transport data can not travel faster than the speed of light. Fiber can move the most significant volume of data at the fastest speeds. Effects to the use of other technologies are driven by augments claiming current data volumes and speeds can be met at lower cost and/or quicker installation times. Many past projects built under these assumptions are now faced with rework because the technology used is now obsolete. Creating the physical link layer with fiber optic technology will support a host of improvements done in software without needing to replace any physical components. Other technologies should be used as a bridge to connect users temporarily while a more ubiquitous fiber optic backbone and last mile infrastructure are constructed.

Wireless technology allows devices within a residence to connect to an access point location only a few feet away. Access points in public places enable devices to connect easily without cables. Wireless technology is best utilized for these types of end device connections rather than network data transmissions over longer distances.

Recommendation: Network infrastructure should be publicly owned and regulated.

Like highways, electric power, and water utilities, public ownership and regulation of broadband infrastructure (fiber optic physical link layer) provide quality services that

© 2022 Research on Knowing LLC

benefit the public interest without monopolies. (POWER, 2019) Public ownership affords the use of transparent processes where the needs and options of all the various community stakeholders can hear and addressed. The resulting open access network can be regulated to benefit the community rather than enrich a few at the expense of the many. The presence of private monopolies within the telecom industry has led to inequalities. The profit motive and requirement for a positive return on investment force private companies to favor high-density population urban areas and neglect sparsely populated rural communities. When service is provided the lack of competitors results in high prices that are often unaffordable.

Recommendation: Decompose the network with a local loop structure to support a divide and conquer strategy driven by support within neighborhoods.

Divide and conquer is a tried and true strategy. Since 1940, rural electrification used Local Utility Districts (LUD) to decompose electric power distribution systems. (Crawford, 2018) A LUD covers a neighborhood of properties. A majority of the property owners within the LUD must agree to participate and pay for the utility buildout. The community interest within a neighborhood along with the financial backing required to successfully petition a Public Utility District (PUD) and form a LUD provides a natural mechanism to prioritize requests based on community interest. The LUD structure has also been used for water utilities. Electric power and water distribution systems are both networks that work with electricity and water respectfully. Broadband can be viewed as just another network that distributes information or data. LUD formation has been used by the Kitsap County PUD in Washington State for Broadband deployment.

A similar decomposition structure to a LUD can be used. For example, Mason County PUD in Washington State uses Fiberhoods. The important characteristics are as follows:

- decomposition of larger network implementation into independent network projects.
- project implementation is driven by community interest and support.

© 2022 Research on Knowing LLC

- projects can be organized into a pipeline or queue for implementation.

The location of projects remains critical to the order of implementation. Projects in more isolated areas must often wait for projects closer to served areas to be implemented. As more projects go online, broadband access is extended making it easier and less expensive to implement future projects.

Additional to the decomposition by residential neighborhoods, building the fiber network to anchor point locations such as schools, fire stations, and other municipal buildings along major corridors enhances the backbone for extending the network to other areas.

The value of identifying smaller neighborhood networking projects where the members of the community are motivated and support broadband implementation is paramount to success. When community engagement is high and the process timeline well understood, each successful project encourages other neighborhoods to participate. As more neighborhoods seek broadband services, the queue of projects enables the public to remain patient as they wait their turn. Each community must know its position within the queue and be informed when projects ahead of them are being implemented.

Recommendation: Multiple methods for project financing should be used.

The typical LUD has each property owner paying for the build-out to their location by dividing the total project cost between the individual participants. Some pay upfront while others use a low-interest multiple-year assessment which is paid with their property owner's property taxes. The usual assessment is for twenty years. Grants from the government and non-profit foundations are another source of financing. Special attention to financing for low-income and non-property owners is vital to insuring equity for everyone needing broadband access.

Internet Service

Recommendation: Internet Service Providers (ISP) acting as retailers should be private sector companies and where necessary public agencies.

© 2022 Research on Knowing LLC

Having a menu of ISPs to select from promotes lower customer prices through competition. Companies can offer a variety of service plans within the regulated standards for quality and service set by the public entity that regulates the broadband infrastructure. Within this scheme, the government agency acting as the regulator for the publicly owned broadband infrastructure operates as a wholesaler to the ISPs within the network. Kitsap County PUD offers users a menu of five ISPs. ("Kitsap PUD," 2022) Each ISP in turn offers a selection of service options.

Each ISP operates under a Service Level Agreement (SLA) with the public broadband regulator. The SLA should specify that ISP services must be net neutral, and meet or exceed network performance & affordability standards. Service should include a month-to-month billing option, and provide symmetrical download/upload speeds with no data limit restrictions. Residential service should be distinguished from business usage and have a separate pricing structure. The distinction between residential and business services rests largely on the bandwidth the service provides along with the data volumes transmitted.

To insure the availability of at least one ISP, the public broadband entity can offer to act as an ISP on the public network they are regulating. Every effort should be made to attract multiple ISP participation. Insuring a low barrier to entry for newly formed ISPs is key to success.

The model of a public network with a menu of ISPs creates an ideal relationship between the public through their governing body and private ISP companies. The strength of government for regulation and transparent public process is maximized. Likewise, private firms can do what they do best. Private companies can innovate ISP services within a publicly owned and regulated network. When properly executed, this model can afford the public the affordable and ubiquitous fiber optic broadband needed to thrive.

Tools and Considerations

© 2022 Research on Knowing LLC

Recommendation: Accurate network performance testing and mapping are critical to both network implementation and management.

Online network performance speed testing tools where users can measure broadband access and share the results with the ISP and the Public Utility are best. The network performance tools must be independently developed and maintained to avoid measurements that inaccurately bias results to the ISP's favor. Measuring short burst data transmissions only provides a best-case performance reading as these types of transmissions are prioritized by the IPS service protocols. The best measurement tools model normal user operations and utilize the multiple communications protocols users employ. Defining normal user operation scenarios is much harder than most people would believe. Internet usage varies greatly from one user to the next. Any measure only creates a single snapshot of network performance. Running the same test repeatedly can yield a range of results. Despite these challenges, testing network performance and mapping the results over time using online crowd-sourced techniques provides critically important information for network implementation and management.

Recommendation: Decisions such as aerial vs buried fiber construction need to be made driven by a cost-benefit analysis of the total cost of service and reliability risks.

While buried fiber construction often provides the lowest total cost of service and highest reliability at the lowest risk, some situations are best served with aerial installation. Only a cost-benefit analysis of the total costs for the lifetime of service can determine the best construction option to use.

Recommendation: Implement Dig Once Policy with incentives rewarding compliance to accelerate network backbone growth and save both time and money.

A Dig Once Policy saves costs and minimizes disruption by avoiding multiple excavations in the same area to install underground conduit with optic fiber elements. By having public works and other construction projects include the installation of conduit with dark fiber for future use, we can accelerate the growth of the broadband

© 2022 Research on Knowing LLC

network and save both time and money. The projects doing this work, however, reap no benefit from the activity. Incentivizing compliance with a Dig Once Policy by creating a reward system would enhance the success of the policy. Projects could be rewarded by estimating the cost and time savings a project creates by the Dig Once Policy. These estimates could be factored into the immediate project's cost and schedule allowing the Dig Once compliance benefits to directly improve the performance measures for the immediate project.

Conclusion

Within this paper, we have presented technical recommendations for creating affordable and ubiquitous fiber optic broadband access. The open-access approach where the public owns and regulates the fiber optic infrastructure and private ISP companies offer end-user services builds on the strength of both government and private enterprise respectfully. The focus has been on providing the essential recommendations informed by best practices from internet technology and project management. We hope this work will be used to create a digital future that includes everyone.

© 2022 Research on Knowing LLC

References

Bode, K. (2022, March 14). Washington State Bill Would Boost Definition Of Broadband To 100 Mbps. Retrieved August 4, 2022, from Techdirt website: https://www.techdirt.com/2022/03/14/washington-state-bill-would-boost-definition-of-broadband-to-100-mbps/#:~:text=Washington%20State%20Senate%20Bill%205715,Mbps%20down%2C%2020%20Mbps%20up.

Crawford, S. (2018, December 11). Fiber: The Coming Tech Revolution—and Why America Might Miss It: 9780300228502: Crawford, Susan: Books.

Kitsap PUD. (2022). Retrieved August 4, 2022, from Broadbandportal.net website: https://kpud.broadbandportal.net/d/Service/ServiceSelect

POWER. (2019, September 3). Public vs. Private: What's Best for Power Customers? Retrieved August 4, 2022, from POWER Magazine website: https://www.powermag.com/public-vs-private-whats-best-for-power-customers/

Velazco, C. (2022, July 19). FCC calls 25 Mbps "broadband" speed. The push is on to up it to 100… Retrieved August 4, 2022, from Washington Post website: https://www.washingtonpost.com/technology/2022/07/19/fcc-broadband-new-definition-100mbps/

Why fiber-optic Internet is a long term, future-proof solution | WRAL TechWire. (2021, July 8). Retrieved August 4, 2022, from WRAL TechWire website: https://wraltechwire.com/2021/07/08/why-fiber-optic-internet-is-a-long-term-future-proof-solution/

© 2022 Research on Knowing LLC

About Authors

Dan Douglass (dan@onKnowing.com) is a contributor to Research on Knowing. He holds a Bachelor of Applied Science (BAS) in Computer Information Systems from Whatcom Community College (WCC). While earning his BAS degree, Dan received a Networking Cybersecurity Associate in Science (AS) degree and Certificates of Proficiency as an Information Security Professional, Cloud Computing, Industrial Control Systems, Mobile Technologies, and Web Development.

Jamie Douglass (jamie@onKnowing.com) is the Principal Scientist and Founder of Research on Knowing. Before establishing Research on Knowing, Jamie worked for 34 years in computing at the Boeing Company where he was an Associate Technical Fellow. Jamie holds a Bachelor of Science (BS) in Computer Science and Mathematics from The Evergreen State College (TESC).

About Research on Knowing, LLC

Research on Knowing pursues the practical application of epistemology to advance how we learn, understand, and create things. Epistemology within Philosophy is "the theory of knowledge, especially with regard to its methods, validity, and scope. Epistemology is the investigation of what distinguishes justified belief from opinion." (OxfordLanguages) We at Research on Knowing believe the practical application of epistemology can open our eyes and change our point of view. The invention of the telescope changed humanity's ideas and understanding of the stars and planets. The microscope made visible an unimagined world within a drop of water. These are examples of how making visible what was previously hidden has revolutionized our knowledge. Computers and software have expanded our knowledge by revealing unseen truths. Research on Knowing employs these kinds of changes in viewpoint to explore how we learn, understand and create things.

© 2022 Research on Knowing LLC